'Assignment-02'

Amirun Nahin

ID: 23201416

Sec: 06-A

Course: CSE350

Date: 28-03-2026

Answer to the question no - 01

(a)

from the given circuit,

$$\text{total resistance in the ladder} = \frac{R}{2} + R + R + \frac{3R}{2}$$

$$= \frac{R + 2R + 2R + 3R}{2}$$

$$= \frac{8R}{2}$$

$$= 4R$$

Now, using voltage divider rule,

$$V_1 = \frac{\frac{R}{2}}{4R} \times (8-0) = \frac{R}{2} \times \frac{1}{4R} \times 8 = 1V$$

$$V_2 = \frac{R + \frac{R}{2}}{4R} \times (8-0) = \frac{3R}{2} \times \frac{1}{4R} \times 8 = 3V$$

$$V_3 = \frac{R + R + \frac{R}{2}}{4R} \times (8-0) = \frac{5R}{2} \times \frac{1}{4R} \times 8 = 5V$$

So, the table listing the quantization intervals:

Quantization Range	Quantization level	Digital output
0-1	0.5	00
1-3	2	01
3-5	4	10
5-8	6.5	11

(b)

The maximum digital input to the DAC is 11 since it is a 2bit DAC.

Now, given that,

$$V_{out} = -6.5$$

$$\Rightarrow -\frac{R_f}{2R}\left(V_{B_1} + \frac{V_{B_0}}{2}\right) = -6.5$$

$$\Rightarrow -\frac{13}{2R}\left(5 + \frac{5}{2}\right) = -6.5$$

$$\Rightarrow -\frac{13}{2R} \times \frac{15}{2} = -6.5$$

$$\Rightarrow \frac{1}{R} \times \frac{-195}{4} = -6.5$$

$$\Rightarrow R = 7.5\ k\Omega$$

We can calculate the step-size from 1LSB voltage,

$$V_{out} = -\frac{R_f}{2R} \times \left(\frac{5}{2} + 0\right)$$

$$= -\frac{13}{2 \times 7.5} \times \frac{5}{2}$$

$$= -2.167\ V$$

So, the resistor's value is 7.5 kΩ and step size is −2.167 V.

Now, output voltages of the remaining inputs are:

input 00:

$$V_{out} = \frac{-R_f}{2R} \times (0+0)$$
$$= 0\ V$$

input 01:

$$V_{out} = \frac{-R_f}{2R} \times \left(0+\frac{5}{2}\right)$$
$$= \frac{-13}{2\times7.5} \times \left(\frac{5}{2}\right)$$
$$= -2.167\ V$$

input 10:

$$V_{out} = -\frac{R_f}{2R} \times (5+0)$$
$$= \frac{-13}{2\times7.5} \times 5$$
$$= -4.33\ V$$

input 11:

$$V_{out} = \frac{-R_f}{2R} \times \left(5+\frac{5}{2}\right)$$
$$= \frac{-13}{2\times7.5} \times \left(\frac{15}{2}\right)$$
$$= -6.5\ V$$

(Ans.)

(c)

from (a) and (b),

digital output for $V_{in} = 5.5V$ is 11.

Now, output voltage,

$$V_{out} = \frac{-R_f}{2R} \times \left(V_{B_1} + \frac{V_{B_0}}{2}\right)$$

$$= \frac{-13}{2 \times 7.5} \times \left(5 + \frac{5}{2}\right)$$

$$= -6.5\ V$$

(Ans.)

(d)

Given,

$n = 2$

$f = 2 \times 10^6$ Hz

We know,

$$t_1 = 2^n T_C$$

$$\Rightarrow t_1 = 2^2 \times \frac{1}{2 \times 10^6}$$

$$\Rightarrow t_1 = 2 \times 10^{-6}\ s$$

$$\Rightarrow t_1 = 2\ \mu s$$

given, conversion time $= 2.7\ \mu s$

$$t_1 + t_2 = 2.7$$

$$\Rightarrow t_2 = 2.7 - t_1$$

$$\Rightarrow t_2 = 2.7 - 2$$

$$\Rightarrow$$

$\Rightarrow \quad t_2 = 0.7\ \mu s$

Now,

$$\frac{V_{in}}{V_{ref}} \times t_1 = t_2$$

$$\Rightarrow \quad V_{in} = \frac{t_2 \times V_{ref}}{t_1}$$

$$\Rightarrow \quad V_{in} = \frac{0.7 \times 8}{2}$$

$$\Rightarrow \quad V_{in} = 2.8\ V$$

and, maximum conversion time $= 2t_1$

$= 2 \times 2$

$= 4\ \mu s$

So, $V_{in} = 2.8\ V$ and $t_{max} = 4\ \mu s$

(Ans.)

(e)

Given, $V_{in} = 5.5\ V.$

Now, we know,

$$\frac{V_{in}}{V_{ref}} \times t_1 = (N+1) \times \frac{1}{f}$$

$$\Rightarrow \quad \frac{5.5}{8} \times 2 \times 10^{-6} = (N+1) \times \frac{1}{2 \times 10^{6}}$$

$$\Rightarrow \quad \frac{5.5}{8} \times 2 \times 10^{-6} \times 2 \times 10^{6} = (N+1)$$

$N+1 = 2.75$

$\Rightarrow N = 1.75$

$\Rightarrow N \approx 1$

$\therefore$ digital output is 01

So, output voltage,

$$V_{out} = -\frac{V_{ref}}{2} \times \frac{R_f}{2R} \times (2b_0 + b_1)$$

$$= -\frac{5}{2} \times \frac{13}{2 \times 7.5}$$

$$= -2.167 \text{ V}$$

$$\therefore \text{Error} = \left| \frac{-2.167 - (-6.5)}{-6.5} \right| \times 100\%$$

$$= 66.67\%$$

(Ans.)

Answer to the question no-02

(a)

from figure 3,

$$R_1 = 2\ k\Omega$$
$$R_2 = 8\ k\Omega$$
$$V_H = 4\ V$$
$$V_L = -8\ V$$

Now,

$$V_{TH} = \left(-\frac{R_1}{R_2}\right) \cdot V_L$$
$$= \left(-\frac{2}{8}\right) \cdot (-8)$$
$$= 2\ V$$

$$V_{TL} = \left(-\frac{R_1}{R_2}\right) \cdot V_H$$
$$= \left(-\frac{2}{8}\right) \cdot (4)$$
$$= -1\ V$$

$$\therefore \quad T_1 = R_i \cdot C_i \left(\frac{V_{TH} - V_{TL}}{V_H}\right)$$
$$= R_i \cdot C_i \cdot \left(\frac{2+1}{4}\right)$$
$$T_1 = \frac{3}{4} R_i C_i$$

and, $T_2 = R_i \cdot C_i \cdot \left(\frac{V_{TL} - V_{TH}}{V_L} \right)$

$= R_i \cdot C_i \cdot \left(\frac{-1-2}{-8} \right).$

$= \frac{3}{8} R_i \cdot C_i$

So, $T = T_1 + T_2$

$= \frac{3}{4} R_i C_i + \frac{3}{8} R_i C_i$

$= \frac{6 R_i C_i + 3 R_i C_i}{8}$

$= \frac{9}{8} R_i C_i$

So, the frequency, we know,

$f = \frac{1}{T}$

$\Rightarrow f = \frac{1}{\frac{9}{8} R_i C_i}$

$\Rightarrow f = \frac{8}{9 R_i C_i}$

(Ans.)

(b)

from 'a',

$$T_1 = \frac{3}{4} R_i C_i$$

$$T_2 = \frac{3}{8} R_i C_i$$

$$\text{Duty Cycle} = \left(\frac{T_1}{T_1+T_2}\right) \times 100\%$$

$$= \left(\frac{\frac{3}{4} R_i C_i}{\frac{3}{4} R_i C_i + \frac{3}{8} R_i C_i}\right) \times 100\%$$

$$= 66.67\%$$

Now, we have to design a square wave generator having duty cycle 66.67%. Given,

$R_1 = 10\ k\Omega$,
$R_2 = 20\ k\Omega$,
$R_x = 10\ k\Omega$,
$C_x = 50\ nF = 50 \times 10^{-9}\ F$
$f = 1\ KHz = 1 \times 10^3\ Hz$

$$\therefore \tau = R_x C_x$$
$$= 10 \times 10^3 \times 50 \times 10^{-9}$$
$$= 5 \times 10^{-4}\ s$$

$$\therefore T = \frac{1}{f} = \frac{1}{1 \times 10^3} = 1 \times 10^{-3}\ s$$

Now, $$T_1 = \tau \ln\left(\frac{V_H - V_{TL}}{V_H - V_{TH}}\right)$$

$$= 5\times10^{-4} \times \ln\left(\frac{V_H - V_L \cdot \frac{R_1}{R_1+R_2}}{V_H - V_H \cdot \frac{R_1}{R_1+R_2}}\right)$$

$$= 5\times10^{-4} \times \ln\left(\frac{V_H - V_L \cdot \frac{10}{30}}{V_H - V_H \cdot \frac{10}{30}}\right)$$

$$= 5\times10^{-4} \ln\left(\frac{\frac{30V_H - 10V_L}{30}}{\frac{30V_H - 10V_H}{30}}\right)$$

$$= 5\times10^{-4} \ln\left(\frac{30V_H - 10V_L}{20V_H}\right)$$

Again,

$$T_2 = \tau \ln\left(\frac{V_L - V_{TH}}{V_L - V_{TL}}\right)$$

$$= 5\times10^{-4} \ln\left(\frac{V_L - V_H \cdot \frac{R_1}{R_1+R_2}}{V_L - V_L \cdot \frac{R_1}{R_1+R_2}}\right)$$

$$= 5\times10^{-4} \times \ln\left(\frac{V_L - V_H \cdot \frac{10}{30}}{V_L - V_L \cdot \frac{10}{30}}\right)$$

$$= 5\times10^{-4} \ln\left(\frac{\frac{30V_L - 10V_H}{30}}{\frac{30V_L - 10V_L}{30}}\right)$$

$$T_2 = 5\times10^{-4} \ln\left(\frac{30V_L - 10V_H}{20V_L}\right)$$

Now,

$$T_1 + T_2 = 1\times10^{-3}$$

$$\Rightarrow 5\times10^{-4}\times\ln\left(\frac{30V_H - 10V_L}{20V_H}\right) + 5\times10^{-4}\times\ln\left(\frac{30V_L - 10V_H}{20V_L}\right) = 1\times10^{-3}$$

$$\Rightarrow \ln\left(\frac{30V_H - 10V_L}{20V_H}\right) + \ln\left(\frac{30V_L - 10V_H}{20V_L}\right) = \frac{1\times10^{-3}}{5\times10^{-4}}$$

$$\Rightarrow \ln\left(\frac{30V_H - 10V_L}{20V_H} \cdot \frac{30V_L - 10V_H}{20V_L}\right) = 2$$

$$\Rightarrow \frac{(30V_H - 10V_L).(30V_L - 10V_H)}{400V_H.V_L} = e^2$$

$$\Rightarrow \frac{900V_HV_L - 300V_H^2 - 300V_L^2 + 100V_HV_L}{400V_HV_L} = e^2$$

$$\Rightarrow \frac{1000V_HV_L - 300V_H^2 - 300V_L^2}{400V_HV_L} = e^2$$

Now, assumming $V_L = -10V$.

$$\Rightarrow \frac{1000.V_H.(-10) - 300(V_H^2) - 300.(-10)^2}{400.V_H.(-10)} = e^2$$

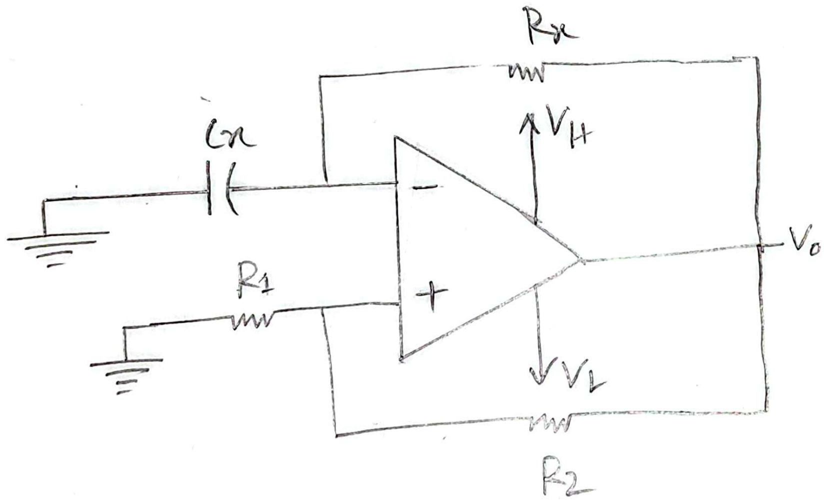
$\Rightarrow -10000V_H - 300V_H^2 - 30000 = -29556.2244V_H$

$\Rightarrow 300V_H^2 - 19556.2244V_H + 30000 = 0$

$\therefore V_H = 1.57V$

So, we get $V_H = 1.57V$ and $V_L = -10V$

Now, the circuit,



$R_x = 10k\Omega$
$R_1 = 10k\Omega$
$R_2 = 20k\Omega$
$C_x = 50nF$
$V_H = 1.57V$
$V_L = -10V$

(Ans.)

Answer to the question no-03

(a)

from 'figure-4',

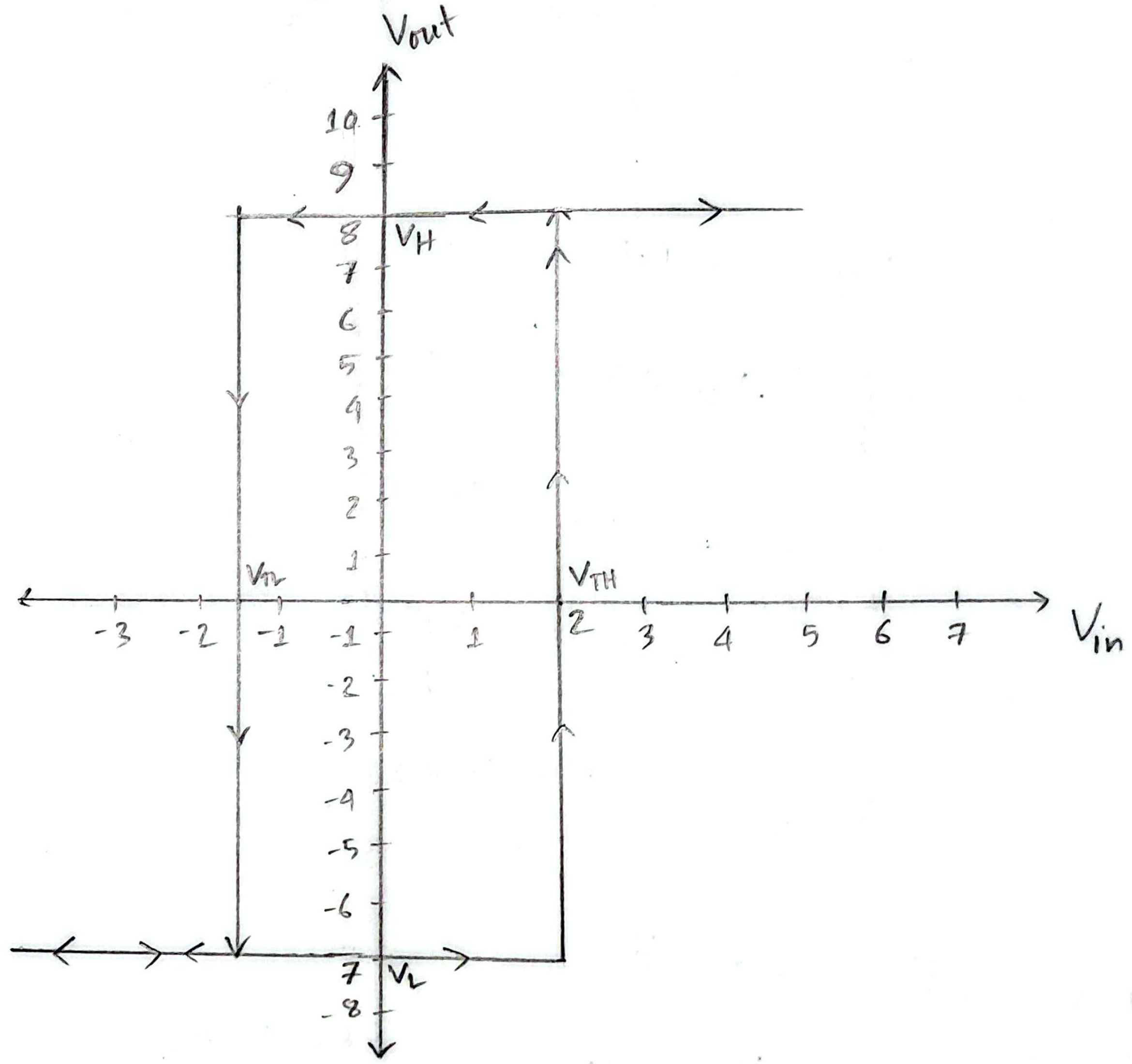
$V_H = 8V$

$V_L = -7V$

$V_{TH} = 2V$

$V_{TL} = -1.5V$

Now, VTC for the given schmitt trigger:



This is a VTC of Non-inverting schmitt trigger since the given schmitt trigger is a non-inverting schmitt trigger.

(b)

from 'a', we got,

$$V_{TH} = 2V$$

$$\Rightarrow \left(-\frac{R_1}{R_2}\right) \cdot V_L + V_S = 2$$

$$\Rightarrow -\frac{R_1}{R_2} \cdot (-7) + V_S = 2$$

$$\Rightarrow V_S = 2 - \frac{7R_1}{R_2}$$

$$\Rightarrow V_S = \frac{2R_2 - 7R_1}{R_2} \text{ ------ (i)}$$

again,

$$V_{TL} = -1.5$$

$$\Rightarrow \left(-\frac{R_1}{R_2}\right) \cdot V_H + V_S = -1.5$$

$$\Rightarrow -\frac{R_1}{R_2} \times 8 + V_S = -1.5$$

$$\Rightarrow V_S = -1.5 + \frac{8R_1}{R_2}$$

$$\Rightarrow V_S = \frac{8R_1 - 1.5R_2}{R_2} \text{ ------ (ii)}$$

from (i) and (ii),

$$\frac{2R_2 - 7R_1}{\cancel{R_2}} = \frac{8R_1 - 1.5R_2}{\cancel{R_2}}$$

$$\Rightarrow 2R_2 - 7R_1 = 8R_1 - 1.5R_2$$

$\Rightarrow \quad 15R_1 = 3.5 R_2$

$\Rightarrow \quad \dfrac{R_1}{R_2} = \dfrac{7}{30}$

$\therefore R_1 = 7\,K\Omega$ and $R_2 = 30\,K\Omega$

Now, for shift voltage,

$$V_s = \frac{V_{TH} + V_{TL}}{2}$$

$$= \frac{2 + (1.5)}{2}$$

$$= 0.25\ V$$

Now, for reference voltage,

$$V_{TH} = \left(-\frac{R_1}{R_2}\right) \cdot V_L + V_s$$

$$\Rightarrow 2 = -\frac{7}{30} \cdot (-7) + V_s$$

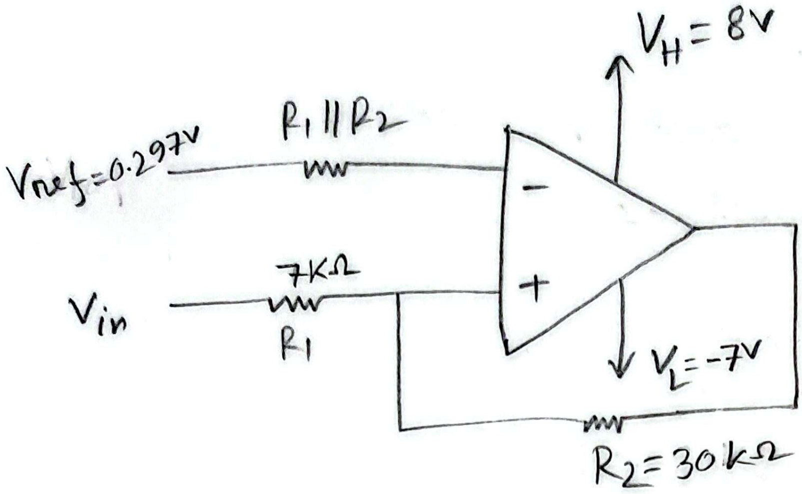
$$\Rightarrow 2 = \frac{49}{30} + \left(\frac{R_1}{R_2} + 1\right) \cdot V_{ref}$$

$$\Rightarrow 2 - \frac{49}{30} = \left(\frac{7}{30} + 1\right) \cdot V_{ref}$$

$$\Rightarrow V_{ref} = \frac{11}{30} \times \frac{30}{37}$$

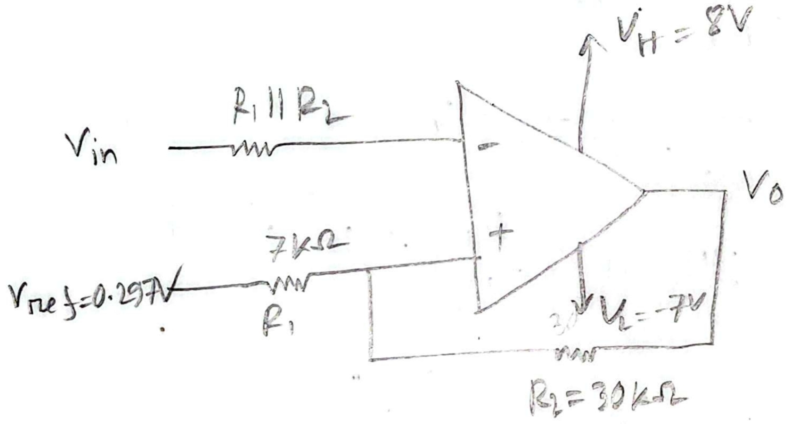
$$\Rightarrow V_{ref} = 0.297\ V$$

Since this is a non-inverting schmitt trigger, the full circuit is drawn below:



(c)

If the connections of input voltage and reference voltages are swapped,



Now, this is a inverting schmitt trigger with reference,

$R_1 = 7k\Omega$

$R_2 = 30\ k\Omega$

$$V_{TH} = \left(\frac{R_1}{R_1+R_2}\right) \cdot V_H + \left(\frac{R_2}{R_1+R_2}\right) \cdot V_{ref}$$

$$= \left(\frac{7}{30+7}\right) \cdot 8 + \left(\frac{30}{30+7}\right) \cdot 0.297$$

$\Rightarrow V_{TH} = 1.754\ V$

and, $V_{TL} = \left(\frac{R_1}{R_1+R_2}\right) \cdot V_L + \left(\frac{R_2}{R_1+R_2}\right) \cdot V_{ref}$

$= \left(\frac{7}{30+7}\right) \cdot (-7) + \left(\frac{30}{30+7}\right) \cdot 0.297$

$= -1.084\ V$

Shift voltage, $V_S = \left(\frac{R_2}{R_1+R_2}\right) \cdot V_{ref}$

$= \left(\frac{30}{30+7}\right) \cdot 0.297$

$= 0.241\ V$

Hysteresis width $= V_{TH} - V_{TL}$

$= 1.754 - (-1.084)$

$= 2.838\ V$.

So, $V_{TH} = 1.754\ V$, $V_{TL} = -1.084\ V$, $V_S = 0.241\ V$ and $V_H = 2.838\ V$ and the modified circuit is inverting schmitt trigger with reference voltage.

(Ans.)